

TManet: An Innovative Approach for Classifying Motor Imagery EEG Signals Related to Motor Movements

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Abstract—The Brain-Computer Interface (BCI) system utilizing Motor Imagery (MI) has diverse applications in medical rehabilitation. This study aims to validate the reliability of self-collected data and classify open-source electroencephalogram (EEG) signals. We introduce a temporal convolution network with multihead self-attention (TManet), designed for decoding EEG signals. This network combined with a data expansion approach based on cyclic translation strategy, which effectively expands experimental data without losing original information. To enhance the processing of self-collected signals, the EEG signals undergo the application of a spatio-temporal filtering technique and a low-rank approximation for generalized eigenvalue decomposition to effectively eliminate multiple artifacts. The proposed model could perform more accurate recognition of self-collected EEG signals than the existing methods, and the highest classification accuracy achieved was 83.75 on self-collected dataset. The effectiveness of this system was proven through the accurate classification of the self-collected MI EEG signals in this study.

Keywords—Brain-computer interface, motor imagery, electroencephalogram, data expansion

I. INTRODUCTION

The Brain-Computer Interface (BCI) is widely used in many industries, such as military, medicine, communication and control [1]. The advancement of this technology primarily relies on the recognition and classification of Motor Imagery (MI) signals. MI based BCI technology is essential for rehabilitating patients with motor disorders like stroke and paralysis, as well as for advancing research in BCI virtual reality devices [2]. Recently various types of physiological data have been incorporated into Brain-Computer Interface (BCI) systems, with motor imagery emerging as one of the most widely utilized non-invasive electroencephalograph (EEG) paradigms. The brain responds to motor imagery in a manner that is correlate with the physical actions that are being imaged or simulated, marked by significant neural activity in the motor cortex. By employing motor imagery decoding, individuals who are physically impaired can effectively control devices and wheelchairs to perform daily tasks, thus proving instrumental in stroke rehabilitation [3].

The accuracy of classifying various EEG signals stands as a fundamental criterion in the assessment of BCI systems. At present, the main methodologies targeted at enhancing the performance of classifying MI-BCI systems revolve around two primary approaches. Refining EEG signals through

preprocessing techniques and optimizing deep learning frameworks are key methodologies employed in enhancing BCI system performance. Over the past few years, deep learning has attracted considerable attention for its exceptional performance in computer vision and natural language processing[4]. Consequently, many researchers have begun to explore how neural networks can be utilized in MI-BCI with excellent results.

In Xu's publication, convolution neural network(CNN) and gradient lifting algorithm were used for feature extraction and classification respectively [5]. It successfully avoided the problem of information loss that occurs during the process of extracting traditional handmade features. The proposed classifier demonstrated an average accuracy of 92% when evaluated on an open-source dataset. An automatic framework based on pre-training CNN, which used multi- scale principal component analysis for denoising and continuous wavelet transform to obtain time-frequency scale maps was proposed by Sadiq et al [6]. The highest average accuracy achieved was 99.52% on the BCI Competition III.

In this study, we present a detailed model that produce improved motor imagery categorization results. Firstly, to remove multi-artifacts, we utilize spatio-temporal filtering and eigenvalue decomposition. Secondly, EEG signal is encoded into high level temporal information using convolutional blocks. Multi-head self-attention(MSA) is applied to emphasize critical information in the time-series [7], followed by a temporal convolutional layer for identifying higher-level temporal features. Additionally, a circular translation strategy is implemented for effective data augmentation. The framework's average classification accuracy for self-collected data and open source data are 83.75% and 89.5%, confirming its effectiveness. In the future, the classification of intricate movements of MI-EEG signals will be enhanced by the foundations laid by this study.

II. MATERIALS AND METHODS

A. Data collection

In this study, MI-EEG signals were collected in an electromagnetic shielding chamber to reduce the influence of the surrounding electromagnetic field and environmental noise. The subjects completed the experiment with comfortable chairs and low light. EEG signals were obtained from 62 scalp sites using Ag/AgCl electrodes, with a Neuro Scan Inc.

Nine healthy volunteers aged between 18 and 26 years were involved. Having normal or corrected-to-normal visual acuity and no history of mental disorders were the expected criteria for participants. They were informed and understood the experimental procedures. The entire data collection was completed in approximately three weeks. Two experimental sessions were performed to collect MI-EEG data from each participant, namely training data and testing data. Each session included 288 trials and was divided into 6 runs. The experimental setup included subjects taking a brief break between successive runs. Movements with the left hand, right hand, both feet and tongue were included in the MI verification tasks. The dataset 2a of BCI Competition IV (<https://www.bbc.de/competition/iv/>) provided by Graz University of Technology was also employed to verify the reliability of our collected MI-EEG data [8].

B. MI-EEG signal preprocessing

The EEG signals underwent downsampling to 250 Hz and filtering via a Butterworth filter, characterized by a center band of ± 1 Hz and a frequency range centered at 0.1 Hz to 50 Hz. The baseline correction was performed to make sure that any epoch was included. Eigenvalue decomposition (EVD) combined with low-order spatio-temporal filtering (STF) removes multiple artifacts from the EEG signal simultaneously. The above steps were implemented using MATLAB and EEGLAB toolbox [9].

The derivation process for removing artifacts is as follows. Suppose the MI-EEG signal $z(t) \in P^M$ Where the sampling time (t) is given in (1):

$$z(t) = x(t) + y(t) \quad (1)$$

$x(t) \in P^M$ is a vector P containing the original neural signal dimension M, and represents the different types of artifacts contained in the original neural signal. Then, the mean subtraction technique is first applied, and the covariance matrix of the signal is expressed as $P_{zz}; P_{xx}; P_{yy}$. Assuming that x and y are not correlated, then a relationship is formulated as in (2):

$$P_{zz} = P_{xx} + P_{yy} \quad (2)$$

The proposed methodology is predicated on a number of underlying assumptions. One of these assumptions is that noise is an additional component of the signal; and the real neural signal and the artificially contaminated signal are not correlated. These are the prerequisites for EEG model. The Wiener filter SFT generates an estimated ($\hat{y}$) of the multichannel artifact signal y through a linear combination of channels z. The estimated signal y is obtained using a linear estimator $\hat{y} = W^T z$, where T above is the conjugate transpose operator. The linear combination is optimized using the mean square error criterion in (3):

$$W^T = \underset{w}{\operatorname{argmin}} E\{\|y - W^T z\|^2\} \quad (3)$$

In fact, the signal correlation matrix P_{zz} is not usually available, but can be estimated using the sample averaging technique. Therefore, the observation matrix Z is defined as

$M \times R$, where the r column represents an observation of Z at a specific time, and the r observations of Z can be separated into a mutually exclusive set using the artifact detection method. Z_a and Z_b , where the first contains artifacts contaminated by R_a , while the second contains artifacts free samples of R_b . This process allows to be estimated as in (4) while the artifact-free sample is estimated as in (5):

$$\hat{P}_{ZZ} = \frac{1}{R_a} Z_a Z_a^T \quad (4)$$

$$\hat{P}_{XX} = \frac{1}{R_b} Z_b Z_b^T \quad (5)$$

By subtracting the estimated artifacts from the addition model using the equation (1), an artifact-free MI-EEG signal is derived from (6):

$$\hat{X} = Z - \hat{W}^T Z \quad (6)$$

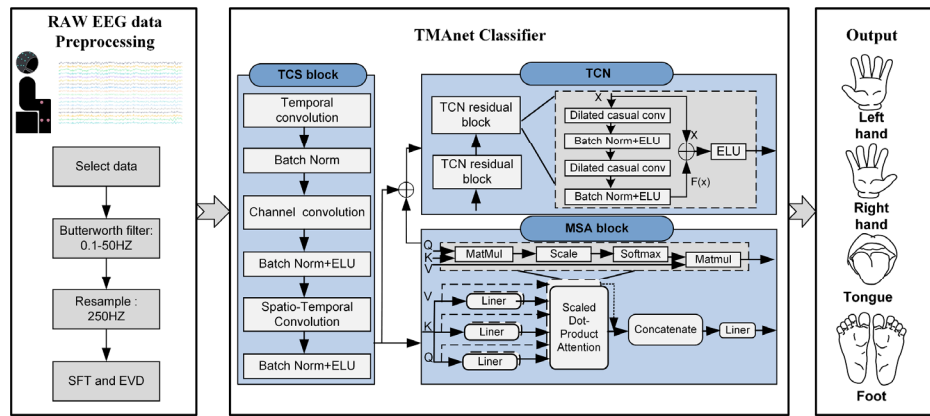
Therefore, the average subtraction process forces to be semi-positive, which enhances the estimation performance of STF, especially under low SNR conditions, and cooperates with the eigenvalue decomposition results.

C. Proposed TManet structure

Recently, the use of deep learning (DL) techniques for decoding MI-EEG brain signals has witnessed a significant expansion, with the convolutional neural network (CNN) being the most widely adopted DL network. This study is achieved by integrating the attention mechanism with DL architectures, allowing for automatic focus on significant elements within input data and layer kernels of MI-EEG data. This study has designed a temporal convolutional network with multihead self-attention (TManet) to classify MI task. It shows in figure 1.

Considering a motor imagery trial denoted as $X_i \in R^{C \times T}$, representing C channels and T time points. In the BCI-2a and self-collected dataset, T equals 1125 time points, with C representing 22 EEG channels.

In the first step, we send X_i to TCS block, which includes three convolutional layers. The initial convolutional layer of the TCS block conducts a temporal convolution on the MI experiment. This convolution uses a filter of size (1, K_C). The value of K_C has been determined to be one quarter of the sampling rate. This facilitates learning spectral features of X_i across various frequency bands and extracting temporal information linked to frequencies above 4 Hz. The second convolution layer employs the technique of channel convolution, which is used to extract spatial features from every feature map. Each convolutional kernel extracts spatial signals from specific frequency bands. Spatio-temporal convolution is the third layer of convolution, which is computed along the temporal direction using a filter of (1, K_{C2}), where K_{C2} is set to 16 for decoding the MI activity within 500ms. To expedite network training, accompanied by batch normalization are the second and third convolutional layers[10]. In addition, the exponential linear unit (ELU) activation is employed to introduce nonlinearity.



In the second step, rather than feeding the entire samples to the subsequent layers, a data augmentation approach is employed to partition the temporal sequences. As shown in the first step, EEG data sample represents as a two-dimensional matrix of $C \times T$. During the data enhancement process, the sample is rotated in the time dimension in step L while keeping the electrode arrangement unchanged. The initial sample range in the time dimension extends from 0 to L. By splicing L to T and 0 to L, the first cyclic translation data is obtained. Similarly, the second cyclic translation data is obtained by splicing 2L to T and 0 to 2L, and so on. In this manner, the K_{th} cyclic translation data is obtained by splicing kL to T and 0 to kL .

In the third step, the MSA block identifies the most valuable feature information in the time series generated by the second step. Performing scaling dot product attention, the multiple self-attention heads make up the MSA. Drawing inspiration from both voluntary and involuntary attention cues, the attention mechanism can be conceptualized through three components: values (representing sensory input), keys (pertaining to involuntary cues), and queries (associated with voluntary cues). The interplay between the query and the key yields an attention score, which in turn directs the selection bias. Afterwards, by linearly projecting the results and concatenating the context vectors of all the heads, the MSA is obtained, which is then added to the input sequence.

Finally, we send the sequence which is from the two step to Temporal Convolutional Network (TCN) block. In a TCN block, a sequence of residual blocks is employed, with each block comprising two dilated causal convolution layers. Following each layer is the application of batch normalization and ELU activation. Dilated convolution enables exponential expansion of the receptive field while increasing network depth. Thus, it is possible to learn the relationships between elements in long sequences through the use of dilation causal convolution. Residual concatenation involves performing. Element-by-element summation of the input and output feature maps, denoted as $F(x) + x$. This technique is applied in residual blocks when identical are the input and output dimensions. For disparate dimensions, linear transformations are utilized, involving a 1×1 convolution. With the increase in the number of stacked residual blocks N , the receptive field size (RFS) of the TCN grows exponentially due to the dilation of each subsequent block. Two parameters, the number of residual blocks N and the kernel size K_T , control the RFS. Defined by the formula $RFS = 1 + 2(K_T - 1)(2^N - 1)$, the receptive field size (RFS) stands at 19 for TCN. Following this, the

outputs of the TCN block from all windows are concatenated. Subsequently, they are forwarded to an FC layer comprising four neurons, aligning with the number of MI classes.

III. RESULTS

A. Results of Preprocessing

Initially, a 0.1-50 Hz band-pass filter was applied to the collected EEG signals to remove high and low frequency noise. Each of the MI trials, with a duration of 4 seconds, underwent sampling at a frequency of 250 Hz. Each trial is associated with one of four MI tasks. It's important to note that there were two separate sessions for each subject on different days. Within each session, subjects underwent a total of 288 trials, thereby furnishing a comprehensive dataset amenable to analysis and comparison. Subsequently, to simultaneously remove multiple artefacts from the EEG signal, we used Eigenvalue Decomposition (EVD) in conjunction with low order Spatio-Temporal Filtering (STF). The proposed method, which exhibits enhanced performance, decreased data demands, and compatibility with limited channels, is a promising avenue for advancing the creation of efficient real-time control strategies for EEG based miniaturized multi-degree of freedom rehabilitation robot interfaces. Figure 2 demonstrates the precision of both the BCI IV 2a dataset and the self-collected data, both before and after processing.

B. Comparison to Recent Studies

This section will present a comparative analysis of the performance of TMA-net with that of other reproduced models,

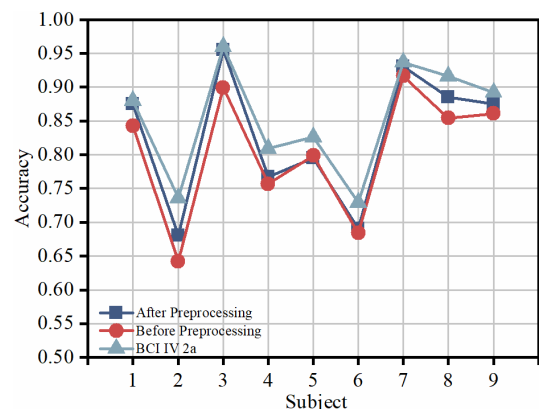


Figure.2 Accuracy comparison of the self-collected and BCI-IV 2a Dataset.

namely EEGnet [11] and CNN[12]. Outperforming recent studies utilizing EEG data is the proposed TMANet model. Compared with EEGnet, utilizing the MSA block to highlight essential information within the temporal sequence characterizes TMANet. Simultaneously, the TCN block is employed to capture advanced temporal features. Additionally, TMANet integrates data augmentation alongside the TCS block to boost the efficiency of MI classification by running the process concurrently.

The training, evaluation and preprocessing of these models followed the methodology outlined in this study, and the parameters defined in the original articles were the basis for the obtained results. TMANet outperformed the other models for all subjects, achieving a highest accuracy of 95.4%, as shown in the figure3. A strong learning capability is indicated by this result from the proposed model, and that it consistently produces similar and stable results over a number of runs. Furthermore, compared to the other models, TMANet demonstrated improved MI decoding for all subjects.

C. Classification Results of TMANet

Figure 4 depicts the experimental results, highlighting the performance of the features extracted by the advanced TMANet model through the survey of the confusion matrices.

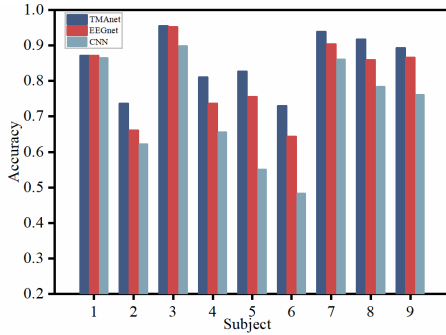


Figure.3 The classification performance of different models.

The results show that when TMANet is used, there is a significant improvement in accuracy across all four Motor Imagery (MI) tasks on open source and self-collected datasets.

While the self-collected dataset demonstrated an overall accuracy of 83.75%, the open-source dataset achieved a higher accuracy of 89.5%. These figures underline the exceptional performance achieved by TMANet across all of the data sets. This underlines the robustness and effectiveness of the model

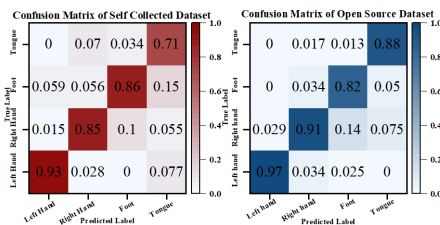


Figure.4 (a) and (b) shows the confusion matrix of two dataset

in achieving consistently high levels of accuracy in the classification of EEG signals, regardless of whether they are obtained from competitions or from self-collected data. This highlights the ability of the model in detecting complex patterns and discriminative features within the EEG signals.

This led to remarkable performance enhancements in the classification of a variety of motor imagery tasks across the study datasets.

IV. CONCLUSION

To sum up, presented in this study is an innovative approach, based on TMANet, that aims to enhance the accuracy as well as generality of Motor Imagery-based Brain-Computer Interface (MI-BCI) systems. Collecting motor imagery EEG signals constituted the basis of the conducted experiments and we would extract various artifacts from these signals. Achieving average accuracies of 83.75% and 89.5% on different datasets, the TMANet model proves to outperform conventional methods. This research not only demonstrates the potential effectiveness of multi-head self-attentive temporal convolutional networks in classifying MI-EEG signals, but also extends the dataset of MI-BCI systems and this approach provides a framework for systems that is compatible with existing standards. The findings of this study are expected to stimulate further research in the field of MI-BCI systems, as they have made a significant contribution to the advancement of this area.

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